

MELISSA SOFIA BINTI SHAHRAN

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SUMMARY

Computer Science undergraduate at UiTM with a CGPA of 3.87 and experience in software quality assurance, web development, and AI-powered systems. Demonstrates strong technical, problem-solving, and leadership skills through professional experience, projects, and competitions. Seeking an internship from **15 March to 18 June 2027**, with flexibility to undertake a longer internship to gain further industry experience.

EXPERIENCES

Pandai Education Sdn. Bhd. **Kuala Lumpur, Malaysia**
Quality Assurance Engineer, Part-Time March 2025 – February 2026
Quality Assurance Engineer, Internship September 2024 – February 2025

- Led 18 testing cycles across Android and iOS applications, comprising 12 regression cycles and 6 release testing cycles.
- Identified and helped resolve 30+ user-reported and functional bugs in collaboration with development and backend teams.
- Supervised and guided QA interns on testing activities, bug reporting, and documentation practices.
- Maintained 3 QA documentation sets covering project timelines, bug-fix logs, and test cases to support testing and release processes.

EDUCATIONS

Bachelor of Computer Science (Hons.) Netcentric Computing **Malacca, Malaysia**
Universiti Teknologi MARA (UiTM), Jasin, Malacca | Current CGPA: 3.87 March 2025 - Present

Diploma in Computer Science **Pahang, Malaysia**
Universiti Teknologi MARA (UiTM), Raub, Pahang | CGPA: 3.91 October 2022 – February 2025
Vice Chancellor's Award | Best Graduate Award

PROJECTS

EcoXchange, AI-Driven Smart Recycling Management System, *Final Year Project | Personal Project*

- Designed and developed a smart recycling management platform using Laravel, Blade, Bootstrap, MySQL and Google Maps API.
- Integrated geolocation and route-related functionality using Google Maps API and Haversine distance calculation, with an intelligent pickup suggestion approach using ACO.
- Designed an AI-assisted recycling item classification workflow using EfficientNet-B0, with Flutter planned for the mobile application.

Project Guardian, AI-powered Maritime Vessel Detection & Situational Awareness System, *SEDIC2026 | 5 members*

- Co-developed a maritime vessel identification system integrating YOLO11 detection and binary classification, achieving 100% selective accuracy (88.9% overall coverage) on an independent 307-image benchmark.
- Designed a fail-closed decision architecture that rejects low-confidence or conflicting predictions, and developed vessel tracking, CPA/TCPA calculations, and a geographic situational-awareness dashboard.
- Built a full-stack operations console with live camera streaming and persistent model inference, supported by 260+ automated tests across frontend and backend.

SmartRoad, Crowdsourced Road Hazard Reporting and Monitoring System

- Led a team of four to develop an Android mobile application to develop a crowdsourced road hazard reporting and monitoring application enabling users to report road hazards and visualize reported locations through Google Maps.
- Built the system using Java, Google Maps SDK, Android Studio, PHP, MySQL and REST API to connect

PROFESSIONAL CERTIFICATIONS

- CompTIA Network+ - CompTIA | February 2026 – February 2029

- Wireless Penetration Tester - 0day | March 2026

ACTIVITIES AND INVOLVEMENTS

- Vice President, SULAM Programme (Service-Learning Malaysia - University for Society), UiTM Jasin 2026
- Student Representative Council (MPP/JPJPP) Election Committee 24/25
- Vice President, *Gerakan Sukarelawan* Club, UiTM Raub 23/24
- Project Manager, Painting Smiles, Creating Hopes (Volunteering Program) 2024
- Secretary, *Ahli Jawatankuasa Pelaksana Parlimen Khas* 2023
- Faculty Facilitator, CalcMastery Tutoring Programme 23/24
- Participated in Building Bridges: Connecting Students with PETRONAS 2023
- Student Representative Council, MRSM Bentong 20/21
- President of Girl Guides Uniform, MRSM Bentong 20/21

COMMUNITY SERVICES

- University representative as Contingent of Unity in Malaysia's Independence Day 2024
- Club representative in *Pusaka Kemanusiaan* at Dong's Village, Raub 2024
- Volunteered as Global Forum International Economic & Finance (GFIEF) 2024 Crew
- Club representative in Painting Smiles, Creating Hope at *Rumah Kanak-Kanak Tengku Ampuan Afzan* 2024
- University representative as *Perpaduan Malaysia MADANI* Contingent *Sempena Sambutan Bulan Kebangsaan* 2023
- Usher VVIP *Majlis Amanat Perdana Hari Kebangsaan Bersama YAB PM* 2023
- Club representative in We Are Family at Rumah Kanak-Kanak Tengku Ampuan Afzan 2023

ACHIEVEMENTS

- 4th Runner Up, Strategic Electronic Defence Innovation Challenge (SEDIC) 2026 in conjunction with CyberDSA 2026
- 2nd Runner Up, 1st International Prosolve 2026, Programming Competition (ICPC)
- Top 10, Road To ICTFF8, Individual 48-Hour Capture The Flag Competition, 2026
- Participant, Taming ICTFF8, Team-Based 12-Hour Capture The Flag Competition, 2026
- 1st Place, Ideathon Innovation Challenge 2025
- Best Graduate Award for the Program & Diploma Program at UiTM Pahang Branch
- Vice Chancellor's Award for Diploma in Computer Science with CGPA of 3.91
- Gold Award, Virtual International Breakthrough Invention, Innovation & Design Exhibition (VIBIIDE) 2024
- 2nd Place, Codeathon, UITM Programming Competition 2024
- 2nd Place, Codevortex Java x C++, Programming Competition 2024
- 3rd Places, Touch Rugby, *Sukan Antara Program* (SUPRO) 2024.
- Bronze Award, International Multimedia Competition 2023
- Bronze Award, International Innovation Competition 2023
- Third Runner-Up Esports Challenge Mobile Legends Bang Bang UITM Cawangan Pahang 2023

TECHNICAL SKILLS

Programming	C++, Java, Python, PHP, JavaScript, TypeScript, SQL, HTML5, CSS, Laravel, Next.js, Flask, Bootstrap, REST APIs, WebSockets
Technologies	PyTorch, TensorFlow, Ultralytics YOLO, EfficientNet-B0, MySQL, Firebase, Git, GitHub, VS Code, Android Studio, Laragon, HeidiSQL, Google Maps API, Leaflet
Multimedia	Figma, Canva, Adobe Photoshop, Adobe Premiere Pro, CapCut, Blender

REFERENCES

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